

INTELLIGENT FEEDBACK SYSTEM

FOR ENHANCING PLACEMENT EXPERIENCES

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CSI4901 Capstone Project

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SYNOPSIS

- Developed an intelligent system for employee review sentiment analysis
- Uses Machine Learning combined with BERT (Transformer Model)
- Text processed using TF-IDF and contextual embeddings
- Compared multiple ML models and deep learning approaches
- Implemented ensemble learning techniques
- Integrated Explainable AI using SHAP & LIME
- Built a web-based interface for real-time sentiment prediction
- Provides aggregated insights across companies, sectors, and locations

INTRODUCTION

- Sentiment analysis is the task of extracting opinions and emotions from textual data
- Employee reviews offer valuable insights into workplace culture and experience
- Manual analysis of large-scale reviews is time-consuming and subjective
- Machine Learning enables automation; BERT improves contextual understanding
- This project develops a hybrid sentiment analysis system with an interactive web interface

SYSTEM OVERVIEW

Input & Preprocessing

- User review entered via web interface
- Text cleaning: lowercasing, stop word removal, punctuation stripping

Feature Extraction

- TF-IDF: term frequency-inverse document frequency vectors
- BERT Embeddings: contextual token representations

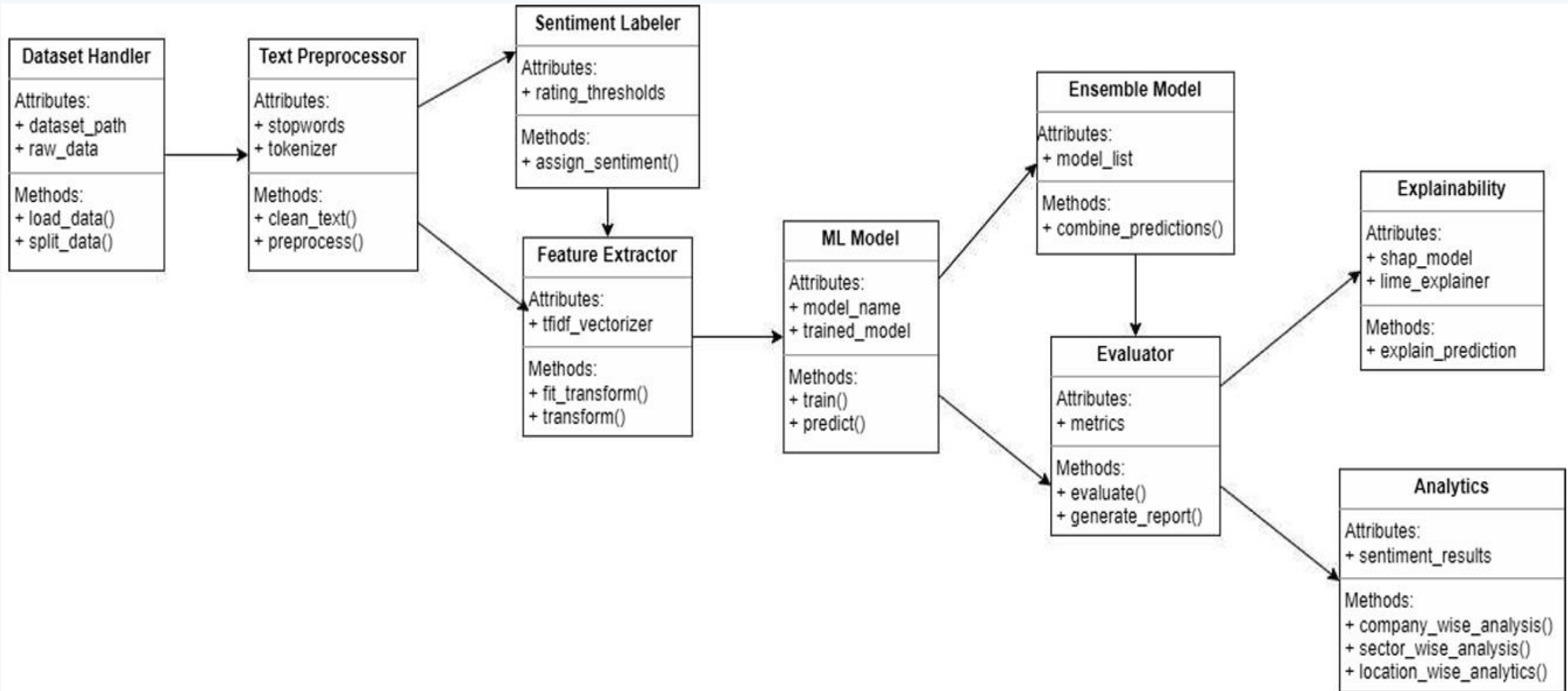
Model Training

- Multiple classical ML models trained and benchmarked
- BERT fine-tuned on domain-specific review data
- Ensemble learning applied for improved accuracy

Output

- Sentiment prediction: Positive, Negative, or Neutral
- Aggregated insights by company, sector, and location

SYSTEM DESIGN DIAGRAM



DATASET DESCRIPTION

Source & Structure

- Dataset: Employee reviews from AmbitionBox
- Contains both structured metadata and unstructured review text

Key Features

- Company Name, Sector, Location
- Employee Size, Review Text, Rating

Preprocessing Steps

- Ratings mapped to three sentiment classes: Positive, Negative, Neutral
- Review text cleaned and tokenized for model input
- Stopword removal, punctuation stripping, lowercasing applied
- BERT tokenizer used for transformer-based feature extraction

IMPLEMENTATION TOOLS & SOFTWARE

Core Language & Libraries

- Language: Python
- Pandas & NumPy — data manipulation
- Scikit-learn — classical ML models
- Transformers (HuggingFace) — BERT model
- SHAP & LIME — Explainable AI

Frontend & Backend

- Frontend: HTML, CSS, JavaScript
- Backend Framework: Flask (Python)

Development Environment

- Jupyter Notebook & Google Colab — experimentation
- VS Code — development & debugging

Results & Performance Metrics

Model: Decision Tree

Accuracy: 0.8020

Confusion Matrix:

```
[[ 1    2    5]
 [ 1 103 234]
 [ 4 150 1500]]
```

Model: Logistic Regression

Accuracy: 0.8275

Confusion Matrix:

```
[[ 0    0    8]
 [ 0    9 329]
 [ 0    8 1646]]
```

Model: Random Forest

Accuracy: 0.8400

Confusion Matrix:

```
[[ 1    0    7]
 [ 0   79 259]
 [ 2   52 1600]]
```

Model: Gradient Boosting

Accuracy: 0.8260

Confusion Matrix:

```
[[ 1    0    7]
 [ 0    4 334]
 [ 4    3 1647]]
```

Model: Multinomial Naive Bayes

Accuracy: 0.8310

Confusion Matrix:

```
[[ 0    0    8]
 [ 0   12 326]
 [ 0    4 1650]]
```

Model: SGD Classifier

Accuracy: 0.8315

Confusion Matrix:

```
[[ 0    0    8]
 [ 0   23 315]
 [ 0   14 1640]]
```

Model: K Nearest Neighbors

Accuracy: 0.8055

Confusion Matrix:

```
[[ 0    0    8]
 [ 0   32 306]
 [ 0   75 1579]]
```

Model: Linear SVC

Accuracy: 0.8320

Confusion Matrix:

```
[[ 1    1    6]
 [ 1   66 271]
 [ 1   56 1597]]
```

Model: SVC

Accuracy: 0.8320

Confusion Matrix:

```
[[ 0    0    8]
 [ 0   25 313]
 [ 0   15 1639]]
```

OUTPUT — INTERACTIVE WEB INTERFACE

What it does

The system exposes an interactive web interface through which users can submit a company name and receive a real-time sentiment prediction based on employee reviews.

Key Capabilities

- Real-time sentiment prediction (Positive / Negative / Neutral)
- Company-level insights aggregated from review data
- Sector and location-based analytics
- Explainability via SHAP & LIME visualizations

The screenshot displays the 'Intelligent Feedback System' web interface. On the left is a sidebar with a logo and navigation links: 'Dashboard' (active), 'Predict Sentiment', 'Company Analysis', and 'Student Feedback'. The main content area is titled 'Input Details' and contains several input fields: 'Company Name' (Intelligent Tech Solutions), 'Company Type' (Startup), 'Sector' (IT Services & Consulting), 'Company Location' (Bangalore, Karnataka), 'Initial Rating (1-5)' (3.8), and 'Placement Experience Review' (New company established in 2023. Good working experience overall). A large purple button at the bottom of the form is labeled 'Analyze Sentiment'. On the right, the 'Analysis Results' section shows a 'Positive' sentiment status, a 'Confidence Score' of 82% (based on ensemble voting model), and 'Key Insights' stating 'Highly positive indicators regarding work environment.' Below this, 'Model Accuracy Metrics' are shown for three models: Hard Voting Ensemble (83%), Random Forest (84%), and Multinomial NB (83%). The bottom of the sidebar shows the user 'Jaishwanth A' with ID '21MID0149'.

Intelligent Feedback System

Dashboard

Predict Sentiment

Company Analysis

Student Feedback

Search companies, sectors, or rev

Input Details

Company Name

Intelligent Tech Solutions

Company Type

Startup

Sector

IT Services & Consulting

Company Location

Bangalore, Karnataka

Initial Rating (1-5)

3.8

Placement Experience Review

New company established in 2023. Good working experience overall

Analyze Sentiment

Analysis Results

Positive

Confidence Score

82%

Based on ensemble voting model (SVC, Random Forest, Gradient Boosting)

Key Insights

Highly positive indicators regarding work environment.

Model Accuracy Metrics

Hard Voting Ensemble	83%
Random Forest	84%
Multinomial NB	83%

Jaishwanth A
21MID0149

CONCLUSION

- Implements BERT to capture contextual meaning of text, outperforming traditional ML approaches such as Naive Bayes and SVM
- Effectively classifies sentiments into Positive, Negative, and Neutral categories for meaningful interpretation
- Web-based interface enables real-time feedback input and instant sentiment predictions
- Transforms unstructured review text into actionable insights for academic institutions
- Demonstrates the potential of combining deep learning with user-friendly interfaces for real-world NLP applications
- Scalable and efficient tool to support data-driven placement strategies and decision-making

FUTURE ENHANCEMENTS

- Multilingual sentiment analysis to support feedback in multiple languages
- Integration of advanced transformer architectures such as RoBERTa or domain-fine-tuned models
- Deployment as a scalable cloud-based API for large-volume real-time analysis
- Voice-based input via speech-to-text technology for improved accessibility
- Expanded and more diverse datasets to improve model generalization
- Advanced analytics dashboards with trend analysis, visualizations, and reporting
- Integration with existing placement portals and college management systems

PROJECT OUTCOME

STTAR - Call for Abstracts | AUI

Ask Gemini

docs.google.com/forms/d/e/1FAIpQLSdtfPqtQULW7VeOcbTps3kjaEqyQ2C9AxUe6EXFiLpt7H8SiA/formResponse

Ask Google

School

All Bookmarks

Gmail

VTOP

Moodle

Perplexity

Entrepreneurship Es...

VIT eGateway

ChatGPT

Teams and Channel...

STTAR - Call for Abstracts | AUI's Annual Interdisciplinary Student Research Conference (2nd Edition, June 2026)

jaishwanth.arun2021@vitstudent.ac.in [Switch account](#) Draft saved

The name, email, and photo associated with your Google account will be recorded when you upload files and submit this form

Any files that are uploaded will be shared outside of the organization they belong to.

* Indicates required question

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THANK YOU

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